

文献阅读笔记

“月船 3 号”月球车坡道：从概念到月面执行

文献信息 Akash A. Shetty et al. “Rover ramp deployment system for Chandrayaan-3: Concept, design, development and operations.” *Acta Astronautica*, 229 (2025), 1–12. DOI: 10.1016/j.actaastro.2024.12.014。

一句话概括 论文完整记录了一个飞行级机构的闭环：受空间包络限制的双面板折叠坡道，通过 SMA Frangibolt 解锁、谐波减速步进电机展开、Kevlar 绳约束月球车下滑，经有限元、热真空、振动、跌落和地面重力补偿试验后，在 2023 年月面成功工作。

设计难题

“月船 3 号”着陆器顶板已被科学载荷占用，26.5 kg 的 Pragyan 月球车只能竖直装在着陆器内部象限，并直接固定在坡道主面板上。坡道因此同时承担两种角色：发射/转移/着陆阶段的承力结构，以及着陆后的展开通道。

月球车静态稳定角约 50° 。为应对着陆点 $\pm 15^\circ$ 地形坡度，作者把坡道名义坡角限制为 30° ，留出约 20° 稳定裕度。铰链离月面约 1250 mm，因此总坡道需约 2500 mm，而发射包络只允许单段不超过 1500 mm，最终采用主、次两面板折叠方案。

概念选择

作者比较两种展开方式：

- 方案 1：次面板由电机先转 180° 并锁定，再由主铰链把主面板、月球车和次面板整体放下。
- 方案 2：主面板电机驱动，次面板依靠重力沿滑轨展开。

方案 2 受滑动摩擦、着陆坡度和局部地形影响，若为不确定性再增加执行器，会提高复杂度。方案 1 的转动过程只要电机有足够重力矩裕度，受地形影响较小，因此被采用。

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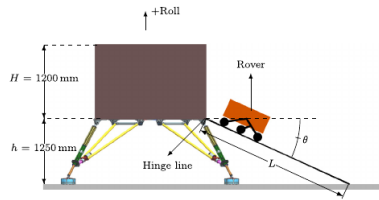


Fig. 3. Configuration of rover ramp.

and facilitates its smooth deployment onto the lunar surface post-landing. Hence, the functional requirements encompass two primary aspects:

2.1. Functional requirement

1. Hold the rover securely, withstanding the dynamic forces during launch, transit to lunar orbit and the lunar descent.
2. Facilitate the controlled roll-out of the rover onto the lunar terrain with slope upto $\pm 15^\circ$.

2.2. Concept

Fig. 3 shows the nominal landing scenario wherein the LM rests on the flat terrain with the ramp in deployed condition. The height of the deployed ramp's hinge line from the lunar surface (along the +Roll axis) measures 1250 mm in nominal condition. The rover schematic is not to scale in all figures that follow.

The length of the ramp (L) is worked out to ensure stability of the rover upon its hold-down release till roll down. Maximum size of ramp along roll axis in stowed configuration is constrained to a maximum of 1500 mm due to launch vehicle envelope. In a single-panel deployment system with $L \leq 1500$ mm the angle θ will be greater than 50° even for nominal terrain condition (Fig. 4). The rover mounted on the ramp is configured with a rocker-bogie based suspension mechanism with a static stability angle of 50° . The angle θ should be less than 35° to ensure rover stability for a terrain slope of 15° at landing site. Thus, $\theta = 30^\circ$ is an adequate choice which demands a total ramp length of 2500 mm.

The friction between wheels and ramp is not sufficient to prevent rover from sliding at 30° angle. A rover wheel restraint mechanism is incorporated in the rover mobility system that prevents the rover from sliding during roll-down. The details are discussed in Section 5.

To accommodate a length of 2500 mm, it is proposed to implement a 2-panel foldable ramp. This design allows the folded ramp to be easily stowed within the lander structure, as illustrated in Fig. 5. The primary panel serves as the platform for mounting the rover and is connected to the LM's bottom deck through a primary hinge line. Similarly, the secondary panel is hinged to the primary panel via a secondary hinge line.

This option optimizes the use of space and ensures efficient stowage of the ramp. Furthermore, the utilization of a 2-panel foldable ramp allows for a seamless transition from stowed configuration to deployment, offering the necessary support for rover operations. The primary panel's pivotal attachment to the lander's bottom deck ensures stability during rover roll-down.

Two options were considered for deploying the panels to form a rigid ramp. In the first option, the secondary panel is deployed first, rotated by 180° using a motor along the secondary hinge line, and

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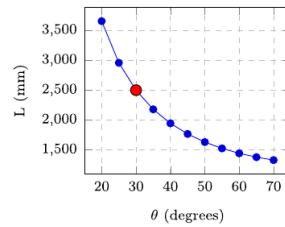
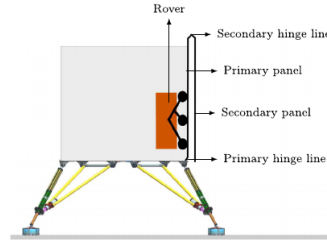
Fig. 4. Variation of the length L with the angle θ .

Fig. 5. Stowed configuration of the rover ramp.

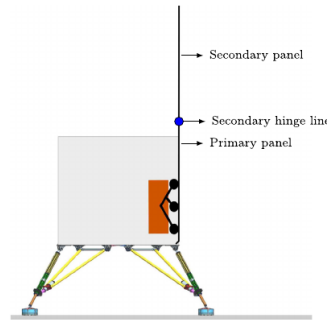


Fig. 6. Rover ramp with secondary panel deployed.

secured with the hinge latch (Fig. 6). Subsequently, the primary panel is deployed along with the already deployed secondary panel, reaching a nominal deployment angle of 120° from the vertical stowed position i.e. $\theta = 30^\circ$.

In the second option, a slider joint is placed between the primary and secondary panels. The primary panel is deployed using a motor along the primary hinge line, while the deployment of the secondary panel is gravity-assisted (Fig. 7).

图：原文第3页。2500 mm 总长与1500 mm 收纳包络的矛盾，直接驱动了双面板构型。

关键子系统

面板与承力路径

主面板为 $1130 \times 870 \times 40$ mm，AA-6061 铝蜂窝/CFRP 面板，质量 4.4 kg；次面板为 $1310 \times 930 \times 15$ mm，质量 1.5 kg。两者均设 CFRP 导轨。月球车以 3 个点固定在主面板，坡道以 4 个保持点固定在着陆器，布置上尽量让坡道保持点对齐月球车保持点，以缩短载荷路径。

保持与释放

4 个非爆炸式 SMA Frangibolt HDRM 在通电加热后伸长并拉断带缺口钛合金螺栓；弹簧提供初始分离。释放顺序是底部 BL/BR → 次面板展开 → 顶部 TL/TR → 主面板展开。接触开关回报释放状态，执行器储存温度必须受控以避免自触发。

驱动与下滑约束

主、次铰链均使用步进电机与谐波减速器。主面板展开经历“主动抬离不稳定平衡点”和“控制重力下放”两个阶段；理论最大需求力矩约 $50 \text{ N}\cdot\text{m}$ ，驱动器通电保持力矩为 $75 \text{ N}\cdot\text{m}$ 。

30° 坡面上的轮—面摩擦不足以阻止月球车滑落，因此后轮缠绕约 2 m Kevlar 绳。车轮滚动时绳索逐步放出，在后轮接触月面前从轮齿槽脱开，实现无需额外主动制动器的速度约束。

分析与地面验证

- 有限元按三个方向分别施加 25 g 准静态惯性载荷；保持点承担 Yaw 方向约 92% 载荷，峰值拔出力 3527 N，对应螺栓预紧力取 5400 N。
- 刚性边界模型估计主/次模态为 61/73 Hz；系统振动试验因着陆器面板柔性，实测首模态约 42.8/60.8 Hz。
- 用氦气球抵消地球重力：次面板所需总升力 43.9 N；主面板连同 22 kg 配重后所需总升力 404.2 N。

- 热真空进行了 -45°C 至 65°C 的 3 个循环，并在 -45°C 浸泡 12 h 后做性能试验；驱动机构无明显退化。
- 原型飞行级正弦振动后 4 个保持点均正常释放；692 mm 跌落试验达到 $2.5\text{ m}\cdot\text{s}^{-1}$ ，前后预紧力下降小于 5%。

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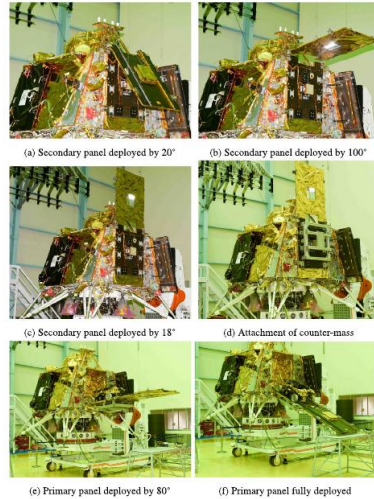


Fig. 28. Sequence of deployment.

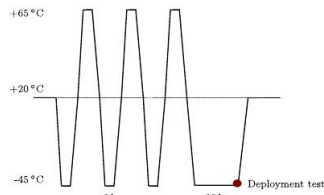


Fig. 29. TVT profile.

consisted of Thermal Vacuum Tests (TVTs) and Vibration Tests (VTs) conducted at both unit and assembly levels. Performance tests were executed before and after these tests to verify system functionality. To verify the performance of the primary and secondary drive mechanisms under extreme temperatures, a test setup was designed to simulate the moment load on the drive hinge lines, equivalent to the expected load during lunar deployment. After achieving a successful vacuum, the setup underwent three thermal cycles between -45°C and 65°C . Performance testing was conducted following a 12 h soak at -45°C of the chamber's base plate Fig. 29. Before initiating motor operations, heaters mounted on the motor casing were activated to achieve the required operating temperature range (0 to 60°C). Both drive mechanisms performed nominally without any degradation.

The QM model, with the stowed RRDS, was subjected to Proto-Flight Level Sine (PFLS) vibration tests. Low-Level Random (LLR) and Low-Level Sine (LLS) signature tests were performed both before and

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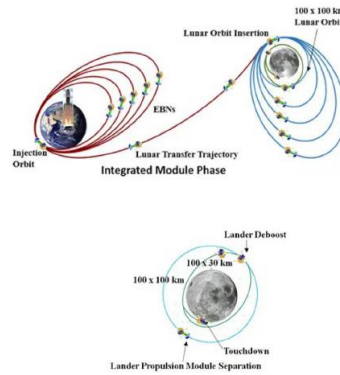


Fig. 30. Mission profile of Chandrayaan-3.

after the PFLS to monitor structural responses and health. LLR tests were conducted prior to LLS tests to establish a notching plan for LLS. An Intermediate Level Sine (XILS) test was conducted before the XPFLS along the X -axis to prevent overtesting due to system non-linearity. From the low-level sine signature tests, the primary and secondary panels' first modes were identified at 42.8 Hz and 60.8 Hz , respectively. These modes were well separated from the structure's and other subsystems' modes and observed to be lower than the subsystem level estimated value due to deck flexibility. Following dynamic testing, RRDS deployment tests were successfully executed with no mechanical anomalies. All four hold-downs functioned nominally. Post-VT, the QM model was subjected to a drop test from a height of 692 mm , achieving a vertical velocity of 2.5 ms^{-1} . The RRDS's integrity was confirmed through preload measurements at the hold-downs, with preload drops remaining below 5% pre- and post-test, indicating system integrity.

Based on the above results, a flight model assembly was developed following the same design concept. The flight model underwent acceptance level tests, including deployment tests, to validate system functionality.

9. Mission phases and telemetry insights

Following the Chandrayaan-3 launch, key mission phases included Earth Burns (EBN), Translunar Injection (TLI), Lunar Burns (LBN), PM-LM separation, Lunar Orbit Insertion (LOI), and two Deboost (DOB) operations before the final powered descent and touchdown (Fig. 30).

Throughout, telemetry data indicated nominal parameters for contact switches/micro-switch on the RRDS. Temperature changes in hold-downs and motors were monitored, with motor temperatures consistently maintained within the storage limit of -45°C to 60°C during maneuvers. Ramp hold-down temperatures, influenced by spacecraft's rotation, were kept within the storage limit of -80°C to 65°C by minimizing direct sun exposure to the $-$ Yaw panel to prevent self-actuation of frangi actuators.

10. Post landing deployment and operations

The ramp mechanism was deployed after nearly 2 hours of landing so as to allow sufficient time for the regolith to settle down and also

图：原文第 10 页，体现从部组件试验到系统环境验证的证据链。

月面实际执行

着陆约 2 h 后，待月尘沉降且热状态合适，系统开始展开。次面板指令 12,000 步，其中先转 150 步确认起动，再转 11,850 步；电位计测得净展开 175.3° ，与地面 176.6° 接近。月面 Frangibolt 动作约 35 s，略慢于室温地面 27–30 s。

主面板转动 17,600 步（ 110° 指令）；电位计在 168 s 内由 2.7° 变至 103° ，取消电机保持后稳定在 122.2° 。月球车随后成功驶出，并完成一个月昼的巡视。

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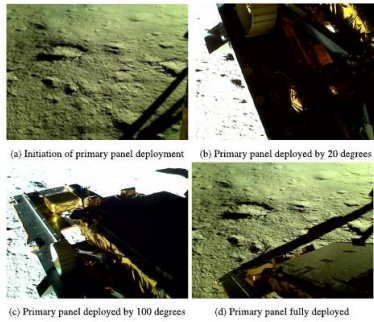


Fig. 31. Deployment of primary panel.

to maintain the thermal requirements aspects of the various systems of the lander etc. The developed mechanism's functioning depends on the successful release of the hold downs and the actuation of deployment motors. The hold-downs and drive modules were covered to protect against lunar dust. The sequence of operation included actuation of hold-downs BL and BR, followed by rotation of secondary panel. Subsequently, TL and TR hold-downs were released followed by primary ramp deployment.

The temperatures of BL and BR actuators were -10°C and 3°C respectively, before heating. The actuators were heated till actuation, which was confirmed by the change in contact-switch status to *Released*. BL's contact switch briefly toggled to *Released* for 3.5 s. Motor dual coil excitation was *Enabled* and the secondary panel was rotated by 150 steps (2.25°) to confirm first motion. BL's release status changed to *Released* at rotation start which was followed by rotation by 11 850 steps (175.3°) for latching. The potentiometer indicated a net deployment of 175.3° which compared well with the on-ground reading of 176.6° . The difference is attributed to the coarse resolution and least count error of 0.97° . The latch status confirmed the successful deployment of secondary panel.

The temperature of TL actuator was -10°C before actuation, which was confirmed by change in micro-switch status. Motor auto-hold was *Enabled* followed by heating of TR actuator which was at 11°C at start of heating. The primary panel was rotated by 17 600 steps (110°). Frangi bolt actuation time for hold-downs on lunar surface was observed to be around 35 s, which is slightly higher than the on-ground observed actuation time of 27–30 s at ambient temperature. The potentiometer angle changed from 2.7° to 103° in 168 s, settling at 122.2° when motor auto-hold was *Disabled*. The motor temperature rose from 20°C to 30°C over 9 min and 14 s during auto-hold *Enable*. Primary panel deployment is shown in Fig. 31. The deployed ramp as seen by the rover is in Fig. 32.

After the deployment of the rover ramp, the rover was successfully rolled out of the ramp. It carried out in-situ exploration of the lunar terrain for one lunar day. The rover motion on lunar surface resulted in few important discoveries. The exploration of the lunar terrain near South pole has led to further understanding of the elemental abundance measurements [24] and the landing site evolution [25].

11. Conclusion

This paper discussed the design, development, and implementation of a rover deployment system onboard Chandrayaan-3. The system

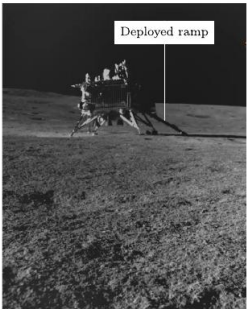


Fig. 32. Lander with deployed ramp.

utilized a foldable ramp mechanism comprising a primary and secondary panel to deploy the rover onto the lunar surface. The rover was affixed to the primary panel, connected to the lander module, while the secondary panel was hinged to the primary panel, both driven by motors along the hinge line. The deployment mechanism incorporated a frangi actuator-based HDRM.

FE analysis was conducted to analyze load distribution among hinges and hold-downs and calculate the required hold-down preload. Extensive qualification tests were performed on a qualification model at both unit and system levels, including deployment tests to ensure system's reliability. Vibration tests confirmed the HDRM and deployable system's resilience during launch, while a cold functional test in a thermo-vacuum chamber validated functionality in near-vacuum conditions. Based on the qualification model's results, a flight model assembly was developed following the same design concept. The flight model underwent acceptance level tests, including deployment tests, to validate system functionality. Following the Chandrayaan-3 lander's successful touchdown, the deployment mechanism operated effectively on the moon's surface.

The foldable ramp mechanism has proven to be a vital element in lunar surface exploration, as demonstrated by its successful deployment during the Chandrayaan-3 mission. The rigorous testing and validation processes undertaken validate the reliability and functionality of the deployment mechanism, affirming its suitability for future lunar and planetary missions.

CRediT authorship contribution statement

Akash A. Shetty: Writing – original draft, Validation, Methodology, Data curation. Harshit Kumar: Visualization, Data curation. Ramprasad Regulavalasa: Visualization. Amit Kamboj: Visualization, Formal analysis. K. Prakasha: Methodology. Raghavendra Prasad H.M.: Supervision, Methodology, Data curation. Gaurav Sharma: Writing – review & editing, Validation, Investigation, Formal analysis. Arun Kumar Gupta: Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. Abhishek Kumar: Writing – review & editing, Supervision. Shankara A.: Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

图：原文第 11 页，左侧为主面板展开，右侧为月面实际部署完成状态。

最有价值的工程经验

- 1. 把承力与部署合并：月球车直接固定在主面板，减少独立承力支架，但也让面板和保持系统承受全任务载荷。
- 2. 采用可观测的顺序机构：接触开关、电位计、温度和电机状态让地面团队能逐步确认再推进。

3. 地面低重力模拟匹配的是“铰链力矩”，不是简单减轻总质量；氦气球连接位置和附加夹具质量都进入力矩平衡。
4. 分析值与系统试验值不一致并非失败。首模态下降揭示了着陆器 deck 柔性，系统级试验正是为捕捉组件边界条件遗漏。

局限与可追问问题

论文证明了单次任务成功，但统计可靠性仍主要来自鉴定流程与设计裕度，无法由一次月面部署给出。文中对月尘罩的结构与密封效果、Kevlar 绳释放的月面遥测、坡面接地状态和月球车实际下行速度披露有限。后续研究可重点比较：多次可复用部署、不同着陆倾角下的接触动力学，以及保持点/电机/传感器的故障容错策略。

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